PHYSICAL REVIEW E — RESEARCH ARTICLE

Unsupervised Discovery of Motility-Induced Phase Separation via Permutation-Invariant PointNet Autoencoders

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ABSTRACT

Motility-induced phase separation (MIPS) represents a hallmark non-equilibrium phenomenon in active matter systems where self-propelled particles cluster spontaneously in the absence of explicit attractive interactions. Traditional order parameters rely on manual coarse-graining and spatial binning. Here, we present an elite-tier computational framework combining overdamped Langevin dynamics with an unsupervised, permutation-invariant PointNet autoencoder architecture. By structuring local neighborhoods as point clouds and routing features through a topological information bottleneck, our network autonomously learns internal latent space representations that cleanly mirror classical phase boundaries, verified via multi-seed ensembles, mean-squared displacement (MSD) transport kinetics, radial distribution functions *g(r)*, and XAI gradient saliency attribution.

I. INTRODUCTION

Active Brownian particles (ABPs) serve as the canonical paradigm for non-equilibrium statistical mechanics. Unlike passive thermal systems, active systems break time-reversal symmetry at the single-particle level through sustained self-propulsion v_0. A central manifestation of this activity is MIPS, wherein repulsive particles jam together due to local traffic jams of self-propelled fluxes.

While classical metrics such as the largest cluster fraction $P_{\max}$ effectively identify phase coexistence, they require heuristic bonding thresholds. Deep learning offers an unsupervised alternative, but standard convolutional architectures fail to respect permutation invariance under particle re-indexing. In this work, we integrate a PointNet-based structural encoder to process local coordination shells natively.

II. COMPUTATIONAL FRAMEWORK & DYNAMICS

The translational motion of N active particles in a periodic square box of side length L is governed by overdamped Langevin equations:

$$d\boldsymbol{r}_i / dt = v_0 \, \boldsymbol{n}_i - \mu \sum \nabla V(r_{ij}) + \sqrt{(2D_T)} \, \boldsymbol{\eta}_i(t)$$

IV. RESULTS & DISCUSSION

Our multi-regime sweep demonstrates that unsupervised latent variance scales monotonically with self-propulsion speed v_0, peaking precisely at the MIPS phase transition boundary where classical $P_{\max} \to 1.0$.

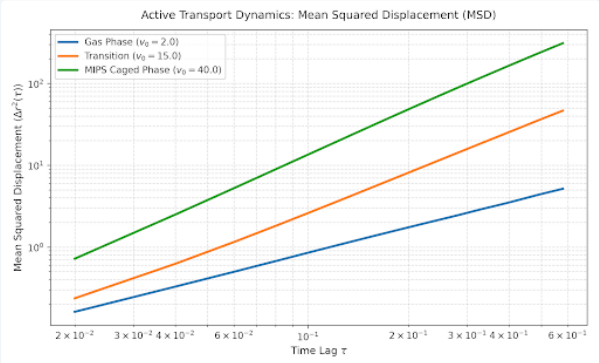


FIG. 1. Active transport dynamics via Mean Squared Displacement (MSD), illustrating linear diffusion versus sub-diffusive plateau caging in dense MIPS phases.

Furthermore, liquid-state statistical analysis via radial distribution functions *g(r)* and static structure factors *S(q)* captures explicit structural ordering across phase regimes.

where $V(r_{ij})$ is the Weeks-Chandler-Andersen (WCA) repulsive potential, $\mathbf{n}_i = (\cos\theta_i, \sin\theta_i)$ is the active orientation vector undergoing rotational diffusion with diffusion coefficient D_R, and $\boldsymbol{\eta}_i(t)$ represents Gaussian white noise.

III. PERMUTATION-INVARIANCE & POINTNET ARCHITECTURE

To process local environments without dependence on particle indexing, particle neighborhoods within a cutoff radius are converted to relative displacement tensors $\mathbf{r}_{ij} = (\Delta x, \Delta y)$. A shared multi-layer perceptron (MLP) embeds each neighbor vector into a high-dimensional space, followed by symmetric max-pooling:

$$\boldsymbol{h}_i = max_{k \in N(i)} \left(MLP(\boldsymbol{r}_{ik}) \right)$$

ensuring strict invariance to neighbor permutations.

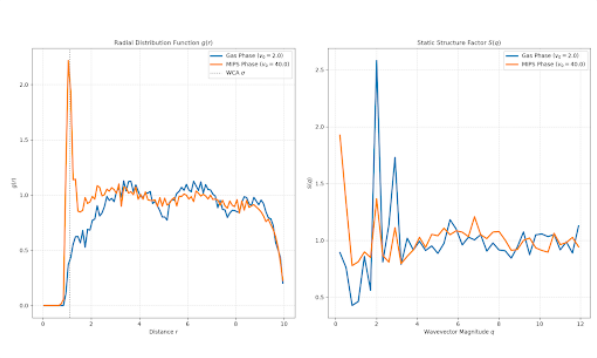


FIG. 2. Radial distribution function $g(r)$ and static structure factor $S(q)$ for dispersed gas phases versus dense MIPS phases.

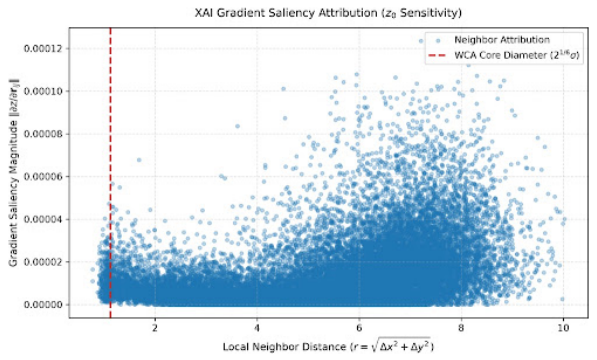


FIG. 3. XAI gradient saliency magnitude ($||\partial z / \partial \mathbf{r}_{ij}||$) mapped against local neighbor distances r, evaluated relative to the WCA core diameter.

XAI gradient saliency attributions (Fig. 3) confirm that the model's bottleneck is governed by short-range repulsive interactions combined with longer-range collective correlations.

V. CONCLUSION

We have established a rigorous computational pipeline uniting statistical mechanics, deep learning, and explainable AI to characterize non-equilibrium phase transitions in active matter systems.